



University of Colombo, Sri Lanka
University of Colombo School of Computing
BACHELOR OF SCIENCE IN COMPUTER SCIENCE

First Year Examination - Semester I - UCSC AY21 [held in Sept / Oct 2024]

SCS1303 — Introduction to Software Engineering

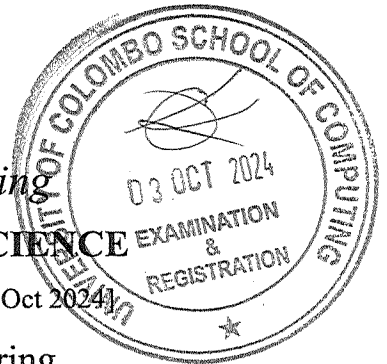
(Two (2) Hours)

Answer ALL questions

Number of Pages = 12

Number of Questions = 3

To be completed by the candidate																	
Index Number:					<table border="1" style="display: inline-table; border-collapse: collapse;"> <tr> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> </tr> </table>												



Important Instructions to Candidates:

- I. Students should answer in the medium of English language only using the space provided in this question paper.
- II. Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- III. Write your index number **CLEARLY** on each and every page of this Question paper.
- IV. This paper consists of 3 questions in 12 pages (including the Cover Page).
- V. Answer **ALL** questions.
- VI. Programmable Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, are **not allowed**.
- VII. Non-Programmable calculators are also **not allowed**
- VIII. Calculators and any electronic device capable of storing and retrieving text, including electronic dictionaries, smartwatches, and mobile phones, are **not allowed**.
- IX. Do not tear off any part of this answer book. Under no circumstances may this book, used or unused, be removed from the Examination Hall by a candidate.

To be completed by the examiners

1	
2	
3	
4	
Total	

Q1. Question 1 has 25 MCQ-based questions.

(2*25=50 marks)

- Each MCQ question will have 5 (five) choices with **one or more correct** answers.
- Answers should be marked on the special answer sheet provided.
- When answering the MCQ questions, completely read and follow the instructions given in the "Instructions to use the MCQ answer sheet" document before you shade your correct choices (The backside of the "special MCQ answer sheet").
- There will be a penalty for selecting wrong choices in a MCQ answer. The maximum mark for an MCQ question is 2 marks, and the minimum mark is 0 marks.

1)

Which of the following is/are examples of embedded software systems?

- (a) Software controlling an anti-lock braking system in a car.
- (b) Word processing software for a desktop computer.
- (c) The operating system on a mobile phone.
- (d) Software that controls the temperature of an Air Conditioner.
- (e) An e-commerce website for online shopping.

2)

Which of the following is/are common causes of software project failures?

- (a) Project objectives are too specific
- (b) Overestimation of resources needed for a project
- (c) Technical terminology used in the communication among stakeholders
- (d) The use of mature technology for development
- (e) Inability of manage the complexity of a project

3)

What is/are not phase(s) of the software development life cycle?

- (a) Feasibility Study
- (b) Software Architecture
- (c) Modeling and Design
- (d) Project Management
- (e) Testing

4)

Which of the following is/are not a plan-driven software development model?

- (a) Waterfall Model
- (b) Prototyping
- (c) Extreme Programming (XP)
- (d) Kanban
- (e) Formal Methods

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5) Which of the following phrases describes (s) the Waterfall model in comparison to other software process models?

- (a) Linear sequence of phases without overlapping
- (b) More suitable for projects with changing requirements
- (c) Early delivery of partial functionality
- (d) Does not emphasize the iterative risk management
- (e) More suitable for small projects

6) Which of the following(s) describes the Spiral Model?

- (a) Emphasizes risk management from the early stage of development
- (b) Each loop in the spiral does not represent the development phase
- (c) Each phase considers the feedback from the previous loop
- (d) Risk assessments are carried out throughout the process
- (e) Not suitable for projects with high levels of uncertainty and risks

7) Which of the following is/are work processes in the Unified Process?

- (a) Inception
- (b) Business Modeling
- (c) Elaboration
- (d) Completion
- (e) Testing

8) Which of the following is/are activities performed using the Scrum methodology?

- (a) Daily stand-up meetings where team members discuss progress and blockers
- (b) Sprint planning to define tasks and allocate work for the upcoming iteration
- (c) Release planning where the entire project is planned in detail before any work begins
- (d) Sprint review meetings to inspect the outcome and gather feedback
- (e) Backlog grooming to refine and prioritize tasks for future sprints

9) Which of the following is/are key practices of Extreme Programming (XP)?

- (a) Pair programming, where two developers work together on the same code.
- (b) Refactoring to continuously improve the code structure without changing its functionality.
- (c) Upfront design before any coding begins.
- (d) Tests are written before writing the actual code.
- (e) Flexible and long working hours.

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10)

What are the primary objectives of the daily Scrum meetings?

- (a) Update team members on each other's progress
- (b) Discuss long-term project planning and budgeting
- (c) Reassign tasks among team members to balance workload.
- (d) Inspect the work completed since the last meeting and plan the work for the day.
- (e) Solve any complex problems or issues that arise during development

11)

Which of the following is/are correct with respect to System Requirements?

- (a) provide detailed descriptions of system functions and constraints.
- (b) written to be understood easily by end-users.
- (c) include both functional and non-functional aspects.
- (d) cannot be linked to user requirements.
- (e) Requirement imprecision is higher in system requirements than user requirements.

12)

Which of the following is/are the types of functional requirements?

- (a) describe the services the system should provide.
- (b) focus on the system's behavior under specific conditions.
- (c) outline system performance and scalability constraints.
- (d) define the interactions between the system and its environment.
- (e) specify how well the system performs its tasks.

13)

Which of the following areas is/are considered as the requirements management?

- (a) Requirement gathering
- (b) Requirement Documentation
- (c) Manage requirement changes
- (d) Requirement Review
- (e) Requirement Traceability

14)

Which of the following is/are the requirement validation techniques?

- (a) System Modeling
- (b) Requirements Reviews
- (c) Test Case Generation
- (d) Requirement Inspection
- (e) Simulation

15)

When selecting a programming language to implement a software product, which factor(s) is/are most important to consider?

- (a) The efficiency and performance of the software to be produced.
- (b) Reputation of the company or group behind the language
- (c) The language's ability to meet system requirements.
- (d) Compatibility with existing systems and libraries.
- (e) Availability of supporting Integrated Development Environment.

16)

Which of the following is/are good coding practices?

- (a) Writing clean and readable code by using consistent indentation.
- (b) Using global variables to store all important data for easy access.
- (c) Keeping functions short and focused on a single task.
- (d) Avoiding comments to reduce unnecessary lines of code.
- (e) Following naming conventions to improve code clarity.

17)

Which of the following is/are correct concerning the Verification and Validation (V&V) Model?

- (a) Verification ensures the product is built correctly according to specifications.
- (b) Validation ensures that the product meets the user's actual needs.
- (c) Verification focuses on building the right product for the customer.
- (d) Validation checks whether the product works in the intended environment.
- (e) Verification and Validation activities are conducted only after implementation.

18)

Consider the following pseudocode.

Read (X, Y);

While (X < Y) X+=2;

Which of the following is/are acceptable test data sets for X and Y?

- (a) (0, 5), (5, 5), (-3, 3)
- (b) (10, 5), (5, 10), (3, 3)
- (c) (0, 0), (-5, 0), (5, 10)
- (d) (-10, -5), (-5, -10), (5, 5)
- (e) (5, -5), (7, 7), (10, 15)

19)

Which of the following statement(s) correctly describes Software Engineering (SE)?

- (a) SE does not consider all aspects of the software development life cycle
- (b) SE is an engineering discipline for both development and maintenance
- (c) SE does not focus on coding practices of software project
- (d) SE fails in large-scale software development projects
- (e) SE is also involved in project management and cost estimation

20)

Which of the following is/are not allowed under Free and Open Source software freedoms?

- (a) Modify the software to create a new version without seeking permission
- (b) Analyze the source code freely without needing authorization
- (c) Distribute the software under a different license without any changes
- (d) Sell the software to others
- (e) Modify the code before executing it

21)

Which of the following statement(s) correctly describes the Iterative and Incremental development process?

- (a) In the Iterative process, the system is developed through repeated cycles, allowing improvements and refinements at each iteration.
- (b) The Incremental process involves breaking down the system into small parts that are developed and delivered sequentially
- (c) Both processes focus on delivering the system without intermediate versions.
- (d) Iterative development focuses on continuous feedback rather than incremental development
- (e) Both processes do not support any changes during the development

22)

Which of the following strateg(y/ies) help(s) in coping with changes during software development?

- (a) Using Incremental development to implement changes in smaller parts of the system.
- (b) Freezing the requirements early in the project to prevent any changes from affecting the development process.
- (c) Adopting Agile methodologies to allow flexibility and continuous adaptation to changing requirements.
- (d) Avoiding customer feedback during development to minimize disruptions caused by new requests.
- (e) Anticipating possible changes and designing software with flexibility in mind.

23)

Which of the following statement(s) is/are correct concerning Development Testing?

- (a) primarily carried out by a separate testing team
- (b) consists of all testing related to unit testing, component testing, and system testing
- (c) concerned with testing the final product before its release
- (d) focuses on identifying and fixing bugs
- (e) carried out before system testing

24)

Which of the following is/are the key benefits of using prototyping in software development?

- (a) improves system usability by allowing early feedback from users.
- (b) eliminates the need for formal documentation and requirements gathering.
- (c) helps clarify vague requirements and identify potential issues early in the process.
- (d) allows customers to experience and interact with the system before the final version is developed.
- (e) guarantees that the final system will be developed faster and with fewer iterations.

25)

Which of the following are key characteristics of a well-written Software Requirements Specification (SRS)?

- (a) defines both functional and non-functional requirements.
- (b) includes the system's design and implementation details.
- (c) serves as a contract between the client and the development team.
- (d) written in a way that all stakeholders, including non-technical users, can understand
- (e) contains detailed test plans for system validation.

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Q2.

- A . Many factors/situations could affect a crisis situation in a software project. The cost overrun is one of the reasons for this situation. What is the expected percentage of software maintenance cost considering the total cost? Draw a graph showing the total cost of a software project and development time.

(06 marks)

ANSWER BOX

Q2.

- B. Briefly explain two ethical responsibilities of software engineers who must uphold during software development.

(06 marks)

ANSWER BOX

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Q2.	C.	Write two (2) functional and one (1) nonfunctional requirement based on the paragraph below. "A Hospital Management System must allow patients to schedule appointments with doctors and view their medical history online, ensuring easy access to healthcare services. It should also enable hospital staff to manage patient records, billing, and inventory efficiently. Additionally, the system must be secure and ensure patient data privacy, complying with healthcare regulations. The system should also have fast response times to handle a large number of users without significant delays."" (06 marks) <u>ANSWER BOX</u>
Q2.	D.	Briefly describe how the incremental model handles changing requirements and give one (1) advantage and one (1) disadvantage of using this model. (07 marks) <u>ANSWER BOX</u>

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- Q3.** A. Which architectural pattern is best suited for software systems where different components run simultaneously and are connected via the Internet? Provide an example of an application that uses this architecture. **(06 marks)**

ANSWER BOX

- Q3.** B. State any **two (2)** values described in the Agile Manifesto. Describe how these values are supported, considering both Scrum and XP methodologies. **(07 marks)**

ANSWER BOX

Index No.

Q3.

C.

A patient can book an appointment with a doctor through the hospital's online system or by calling the hospital directly. When booking online, the patient must log in using their credentials. New patients can sign up to create an account and obtain login credentials. The system allows patients to save their appointment requests as drafts before confirming the appointment.

Draw the Use Case Diagram for the above scenario, incorporating generalization, including and extending relationships. (06 marks)

ANSWER BOX

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Q3. D.

A patient can book appointments with multiple doctors. However, a doctor's appointment is only confirmed if at least three patients have booked appointments for that doctor on the same day. Some appointments may require prior medical tests or consultations as prerequisites. Patients can either be in-patients or out-patients. Staff members are either doctors or nurses, but some staff members serve dual roles as both doctors and nurses.

Draw the class diagram for the above scenario.

(06 marks)

ANSWER BOX

*****End of the Paper *****